Investigating EEG correlates of speech segmentation in Basque young bilinguals: the role of individual differences

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The ability to segment continuous speech into discrete linguistic units is a primary research interest in psycholinguistics, as successfully parsing speech is a prerequisite for accessing lexical and grammatical structure. Electrophysiological (EEG) studies investigating speech segmentation have shown that word-onset syllables elicit larger (i.e., more negative) N1 and/or N2 amplitudes than acoustically matched medial syllables, indexing online word-boundary detection (Sanders & Neville, 2003a,b). The extent and distribution of these effects also varies with language experience, with native speakers typically exhibiting stronger and more consistent word-onset effects than non-native speakers. Here, we investigate whether degree of bilingual experience modulates word-onset effects (i.e., N1 and/or N2 amplitudes) in native and non-native speakers of Basque. Specifically, we examine the effects of age of acquisition (AoA), proficiency, language use, and language switching frequency, measured continuously.

Participants were 124 Basque-Spanish bilinguals from the Basque Country (97 females, M=21.31, 18–35), including 37 native and 87 non-native speakers of Basque. Bilingual experience was assessed using a combination of self-report- and objective measures. EEG was continuously recorded while participants completed a segmentation task in Basque, adapted from Sanders and Neville (2000), which comprised 240 sentences of three types: *Natural* sentences, i.e. fully grammatical and meaningful; *Jabberwocky* sentences, where content words were replaced with nonsense words, but functors were preserved (i.e., grammatical, but not meaningful); and *Phonological* sentences, comprising only nonsense words, but preserving all phonological information (i.e., neither grammatical, nor meaningful). Each sentence contained a target phoneme and the participant's task was to indicate whether it occurred word-initially or -medially (henceforth WI and WM).

To explore the modulation of the four bilingual factors, we created group-specific predictors for each of them, containing all tasks used in their assessment, i.e., self-reported AoA and individual principal components—derived from PCAs—for proficiency, use, and switching measures. In the case of natives, language use and switching patterns were highly correlated (r=.79, p≤.001), so we performed a combined PCA including all measures from both factors, while in non-natives we kept an individual component for each factor. We first ran cluster-based permutation tests (CBP) separately for each group to identify N1 (96–156 ms) and N2 (200–300 ms) word-onset effects within the 0–300 ms time window (Figure 1). We then fitted linear mixed-effects models to N1/N2 amplitudes averaged over the identified significant clusters, or over the full component time window when no clusters emerged. In natives, N1 amplitudes showed significant left-lateralized negativity (p=.024). Higher language use and switching was associated with more negative N1 amplitudes at anterior sites (p=.030), for WM compared to WI targets (p=.010), and in Phonological compared to Natural sentences (p=.025). The same interaction of sentence type by use and switching was found in the analysis of the N2 (p_s≤.001). In non-natives, sentence type interacted with both language use and language switching in the analysis of N2. Higher language use yielded less negative amplitudes in *Jabberwocky* than in *Natural* and *Phonological* sentences (p_s≤.019). In contrast, higher language switching yielded more negative amplitudes in *Jabberwocky* than in *Phonological* sentences (p=.018). Overall, our results suggest that dynamic patterns of language use and switching (i.e., language experience), rather than language skill (as indexed by AoA and/or proficiency), shape the magnitude and distribution of N1 and N2 word-onset effects in both natives and non-natives.

References

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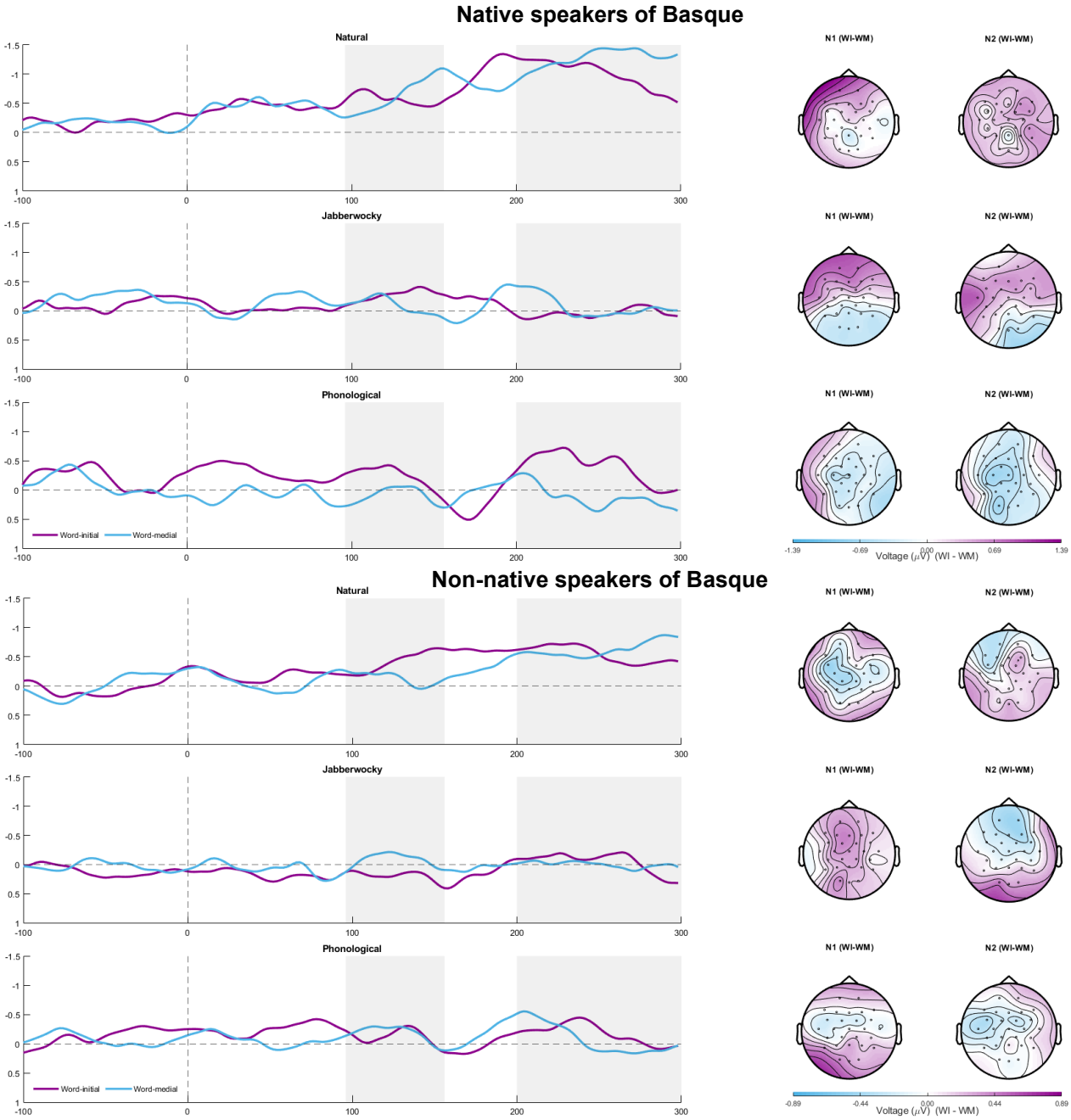

Figure 1. Grand averaged ERP responses over centro-parietal electrodes in native speakers (upper panel: Cz, CP1, CP2, P3, P4) and over fronto-central electrodes in non-native speakers (lower panel: FC5, C3, CP1, CP2, Cz). Left panels show ERP waveforms (purple: word-initial; light blue: word-medial). Right panels display scalp topographies of the difference between word-initial and word-medial conditions (WI–WM) for the N1(left) and N2 (right) time windows.